RV College of Engineering
DEPARTMENT OF INDUSTRIAL ENGINEERING AND MANAGEMENT



III Semester Project Based Learning
IM234AI - MANUFACTURING PROCESS LABORATORY

FLYWHEEL AND CRANK MECHANISM

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CERTIFICATE

This is to certify that the project work titled **"Flywheel and Crank Mechanism"** carried out by **Vivek Bainsla – 1RV24IM061, Aryan Lodha – 1RV24IM064, Yashveer Singh – 1RV24IM061**, in partial fulfillment of the requirements of the **Manufacturing Process Laboratory** course (**IM234AI**) introduced in the curriculum during the **2025–26**, has been completed successfully. It is further certified that all necessary corrections and suggestions indicated during internal assessments have been duly incorporated into the final report.

The project work has been evaluated and approved as it meets the academic standards and experiential learning objectives of the **Manufacturing Process Laboratory** as prescribed by the institution.

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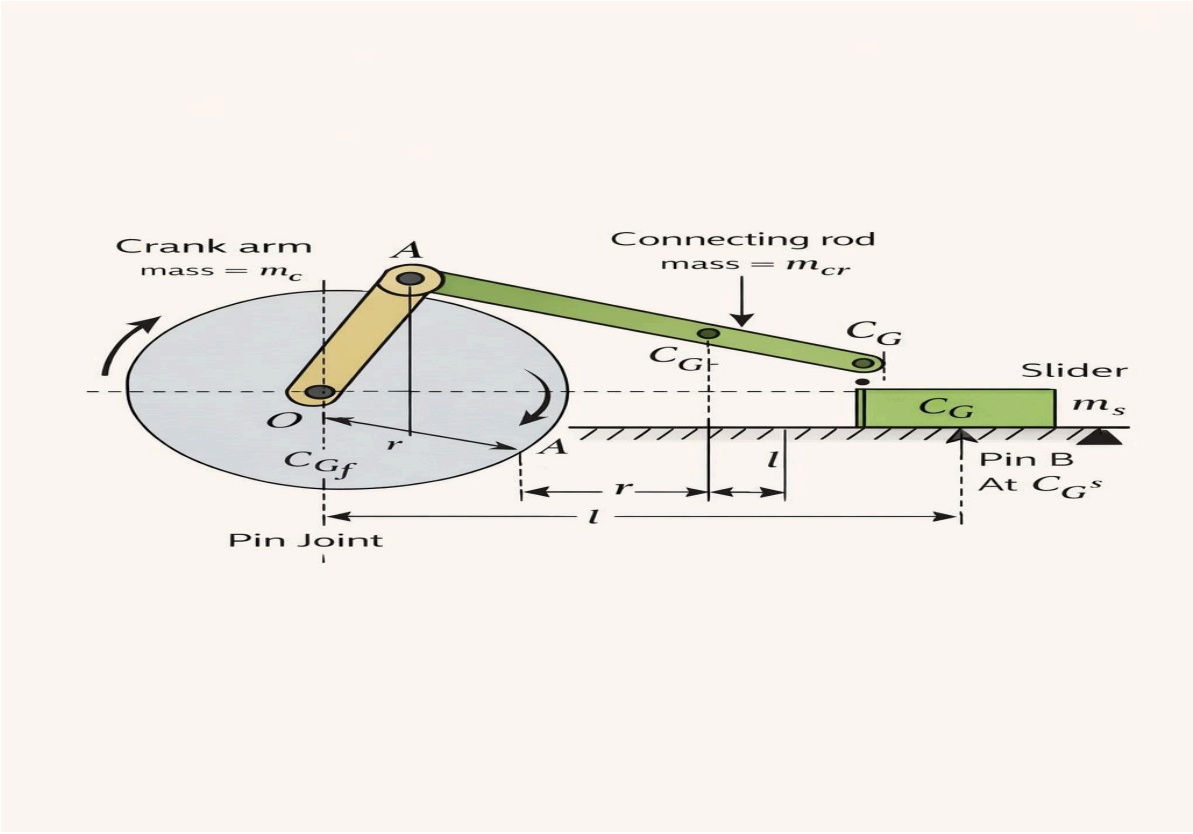
CHAPTER 1: INTRODUCTION

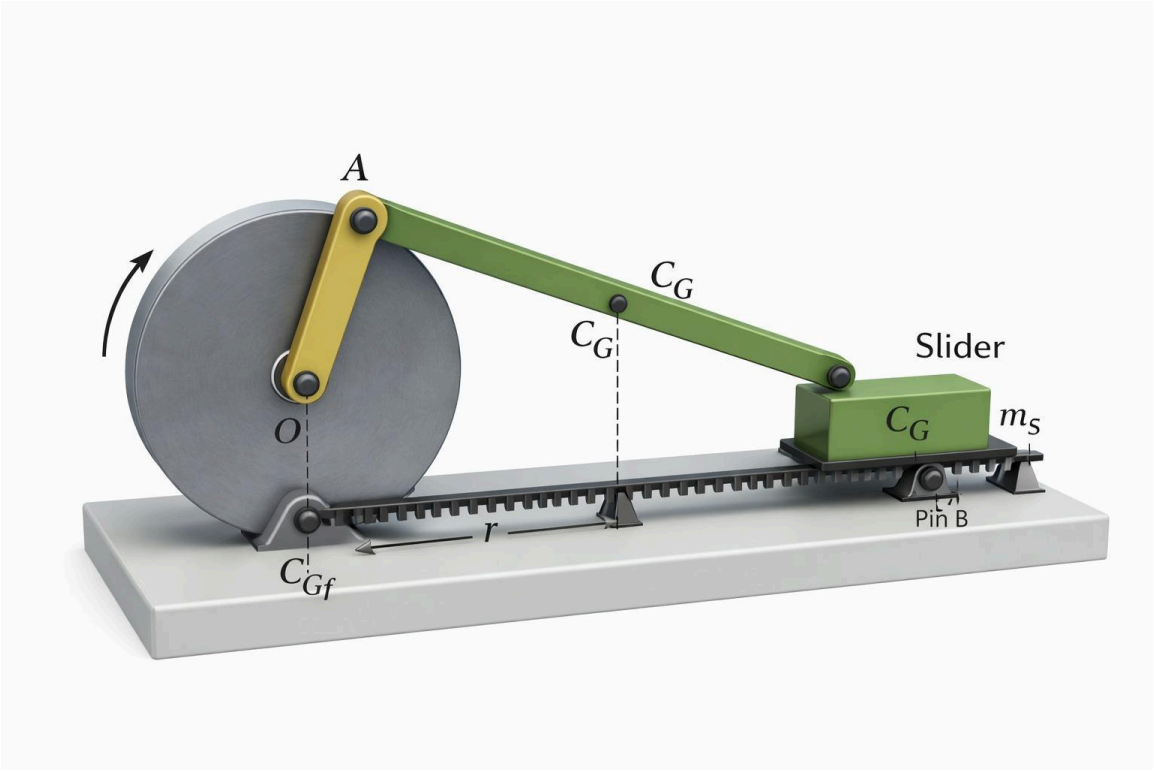
As part of our Project Based Learning in the Manufacturing Process Laboratory, we successfully designed and fabricated a flywheel and crank mechanism using wood, employing both a lathe and a milling machine. The flywheel and crank mechanism is a fundamental mechanical system widely used to convert reciprocating motion into rotary motion or vice versa. It is commonly applied in internal combustion engines, pumps, compressors, and various machine tools where smooth and continuous power transmission is required.

The project aimed to enhance our practical understanding of machine tools, material behavior, and manufacturing processes by transforming theoretical concepts into a functional mechanical assembly. Wood was deliberately selected as the work material due to its ease of machining and its ability to clearly demonstrate the working principle of the mechanism. Operations such as turning, facing, drilling, slot cutting, and dimensional finishing were carried out on the lathe and milling machine, providing hands-on experience in tool setup, machining accuracy, and safety procedures.

Through this project, we developed essential skills in mechanism design, precision manufacturing, teamwork, and problem-solving. Fabrication of the flywheel and crank mechanism offered valuable insights into balance, alignment, inertia effects, and smooth motion transmission, which are critical considerations in real-world mechanical systems. Overall, this project strengthened our understanding of manufacturing processes and reinforced the significance of practical learning in engineering education.

CHAPTER 2: 3D AND 2D DRAWINGS





CHAPTER 3: TOOLS AND MATERIALS USED

1. Power Hacksaw / Band Saw

A power hacksaw or band saw was used for cutting the raw material into required lengths before machining. This machine helped in removing excess material and preparing workpieces of suitable size for further operations. It ensured straight and accurate cuts, reducing manual effort and saving time.

2. Lathe Machine

The lathe machine was the primary machine tool used in the fabrication of the flywheel and crankshaft mechanism. It was used for turning, facing, and finishing operations on cylindrical components such as the flywheel rim, hub, and crankshaft journals. The lathe ensured concentricity, uniform diameter, and smooth surface finish, which are critical for balanced rotation and smooth motion.

3. Drilling Machine

A drilling machine was used to produce holes for bolts, pins, and shafts in various components of the mechanism. Different sizes of drill bits were used depending on the required hole diameter. Accurate drilling ensured proper fitting of fasteners and secure assembly of the flywheel and crankshaft components.

4. Milling Machine

The milling machine was used to machine flat surfaces, slots, and profiles on parts such as the crank arm and mounting surfaces. Using multi-point cutting tools, the milling machine provided high dimensional accuracy and proper alignment of components, which is essential for smooth motion transfer.

5. Grinding Machine

A grinding machine was used for finishing operations to achieve fine surface finish and close tolerances. Grinding reduced friction between moving parts and improved wear resistance, thereby increasing the operational life and efficiency of the mechanism.

6. Bench Grinder
A bench grinder was used for deburring, sharpening cutting tools, and removing sharp edges from machined components. This improved safety during handling and ensured clean edges for assembly.

7. Measuring Instruments
Measuring instruments such as vernier calipers, micrometers, steel rules, and dial gauges were used to check dimensions, concentricity, and alignment of the components at different stages of machining. Accurate measurement ensured dimensional correctness and quality control.

8. Assembly Tools
Hand tools such as spanners, Allen keys, screwdrivers, hammers, and clamps were used during the assembly process. These tools helped in proper fastening, alignment, and secure fitting of all components.

9. Lubrication Equipment
Lubricating oil and grease were used during machining and assembly to reduce friction, prevent overheating, and ensure smooth operation of moving parts.

CHAPTER 4: APPLICATIONS

The flywheel and crank mechanism is widely used in applications where smooth power transmission, energy storage, and conversion between rotary and reciprocating motion are required. One of its most common applications is in internal combustion engines, where the crank mechanism converts the reciprocating motion of the piston into continuous rotary motion, while the flywheel stores excess energy and reduces speed fluctuations. This ensures uniform rotation of the crankshaft, improves engine efficiency, and minimizes vibrations during operation.

Another important application of the flywheel and crank mechanism is in machines requiring periodic force and smooth motion, such as reciprocating pumps, compressors, and presses. In these systems, the flywheel helps maintain momentum during the non-power strokes, enabling steady operation and reducing the load on the prime mover. The crank mechanism ensures accurate and controlled motion transfer, making it suitable for applications where consistent stroke length and timing are essential.

The flywheel and crank mechanism is also extensively used in machine tools and industrial equipment, including lathes, punching machines, and rolling mills. In such applications, the flywheel improves speed regulation and energy efficiency, while the crank mechanism provides precise motion control. Its simple construction, reliability, and ability to handle variable loads make it ideal for continuous and heavy-duty operations. Overall, the flywheel and crank mechanism remains a vital mechanical system in industries where smooth operation, reliability, and effective motion control are critical requirements.

CHAPTER 5: OPERATIONS PERFORMED

1. Cutting Operation:

The cutting operation was the first step in the manufacturing process of the flywheel and crankshaft mechanism. Raw materials such as mild steel rods and plates were cut to the required approximate lengths using power saws or hand cutting tools. This operation helped in removing excess material and producing workpieces of manageable size for further machining processes. Proper cutting ensured minimum material wastage and reduced machining time in subsequent operations. Accurate cutting also contributed to maintaining uniformity in the dimensions of the components before detailed machining.

2. Turning Operation:

Turning was carried out on a lathe machine to machine cylindrical and concentric surfaces of components such as the flywheel rim, hub, and crankshaft journals. During this operation, the workpiece was rotated while a single-point cutting tool removed excess material to achieve the desired diameter and surface finish. Precise turning was essential to ensure uniform rotation, proper balance, and smooth transmission of motion in the mechanism. Any inaccuracy in turning could result in vibration, misalignment, or uneven wear during operation, affecting overall performance.

3. Facing Operation:

Facing operation was performed to produce flat and smooth end surfaces on the flywheel, crankshaft, and other rotating components. In this process, material was removed from the ends of the workpiece to achieve accurate length and perpendicularity with respect to the axis of rotation. Proper facing ensured correct seating of parts during assembly and helped maintain alignment between different components. This operation was crucial for achieving dimensional accuracy and preventing axial play in the assembled mechanism.

4. Drilling Operation:

Drilling was used to produce holes required for shafts, bolts, pins, and fasteners in various components of the flywheel and crankshaft mechanism. The operation was carried out using a drilling machine with appropriate drill bits selected based on the required hole diameter. Accurate drilling ensured proper fitting of fasteners and smooth assembly of the crank, connecting rod, and flywheel. Incorrect drilling could lead to loose fittings, misalignment, or reduced structural strength of the assembly.

5. Milling Operation:

Milling operation was employed to machine flat surfaces, slots, and profiles on components such as the crank arm and mounting areas. Using a milling machine, multi-point cutting tools removed material with high accuracy to achieve the desired shape and dimensions. Milling ensured proper alignment of components and facilitated smooth motion transfer between the crankshaft and connecting rod. This operation required careful setup and precise control of cutting parameters to maintain dimensional accuracy and surface quality.

6. Grinding / Finishing Operation:

Grinding and finishing operations were carried out to improve surface finish and achieve close dimensional tolerances. These processes helped in removing minor surface irregularities left after turning and milling operations. Proper finishing reduced friction between moving parts, minimized wear, and enhanced the smooth operation of the mechanism. A good surface finish also increased the service life and reliability of the flywheel and crankshaft assembly.

7. Assembly Operation:
After completing all machining processes, the individual components were assembled to form the flywheel and crankshaft mechanism. Assembly involved fitting the flywheel onto the shaft, connecting the crank arm, and aligning the connecting rod. Careful attention was given to alignment, fastening, and clearance between moving parts. Proper assembly was essential to ensure smooth rotation, efficient power transmission, and safe operation of the mechanism.

8. Inspection and Testing:

The final step involved inspection and testing of the assembled flywheel and crankshaft mechanism. Dimensional checks were performed using measuring instruments to verify accuracy and alignment. The mechanism was then operated under no-load conditions to observe smoothness of motion, vibration, and noise. Any defects or misalignments identified during testing were corrected to ensure reliable and efficient performance of the system..

PART	DIMENSIONS	OPERATION
FLYWHEEL	**RADIUS: 100mm** **THICKNESS: 18mm**	**1. CUTTING** **2. FACING** **3. TURNING** **4. DRILLING**
CONNECTING ROD	**LENGTH:330mm** **THICKNESS :20mm**	**1. CUTTING** **2. FACING** **3. TURNING** **4. DRILLING** **5. MILLING** **6. GRINDING**
SLIDER	**LENGTH:330mm**	**1. CUTTING** **2. FACING** **3. TURNING** **4. DRILLING** **5. MILLING** **6. GRINDING**
SLIDER BLOCK	**HEIGHT:40mm** **LENGTH :48mm** **BREADTH:36mm**	**1. CUTTING** **2. FACING** **3. TURNING** **4. DRILLING** **5. MILLING** **6. GRINDING**

CHAPTER 6: CONCLUSIONS AND PERSONAL REFLECTIONS

CONCLUSIONS

The fabrication of the flywheel and crank mechanism using wood successfully demonstrated the practical application of manufacturing processes such as facing, turning, and milling. This project enabled us to convert theoretical concepts of rotary and reciprocating motion into a functional physical model, highlighting the importance of dimensional accuracy, alignment, and proper machining practices. The completed flywheel and crank mechanism operated smoothly and effectively transmitted motion between components, thereby validating the design methodology and manufacturing approach adopted during the project.

PERSONAL REFLECTIONS:

Working on this project was a valuable learning experience that enhanced our confidence in operating machine tools and understanding real-world manufacturing challenges. It improved our skills in planning machining operations, maintaining dimensional accuracy, and assembling mechanical components with precision while working collaboratively as a team. This hands-on experience not only strengthened our technical knowledge but also emphasized the importance of safety, patience, and attention to detail in engineering practice. Overall, the project made the learning process more engaging and reinforced our interest in practical mechanical design and manufacturing.

CHAPTER 7: VISUALS





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